Technische Hochschule Augsburg

STATISTICS

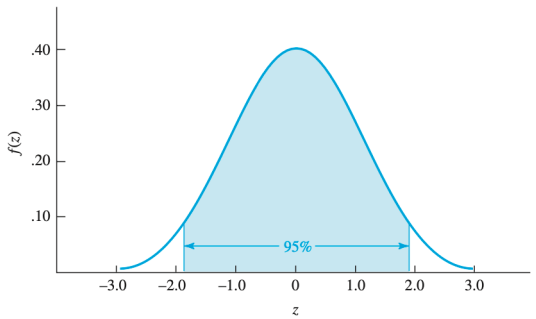
International Information Systems, 3rd semester

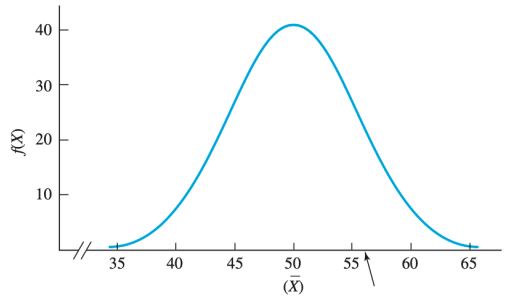
(9) Test Statistics (2 & 3)

PROF. DR. -PHIL ALESSANDRA ZARCONE

LAST WEEK

Test statistics





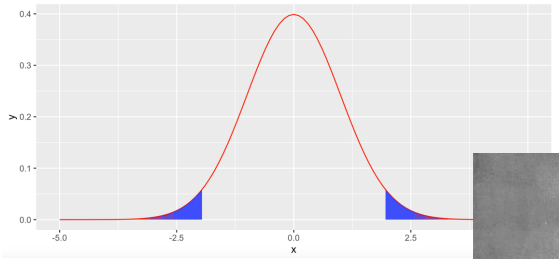
- For a **single score**:
  - **Probable Limits** indicating where scores may likely fall in a distribution
  - **Area under the Curve** to calculate the probability that a score comes from a population
- For a **sample**:
  - **Confidence Intervals** where we expect to find the population parameter
  - **Area under the Curve** to calculate the probability that the sample comes from a sampling distribution
- What is a good **rejection level** to **reject H_0**? It depends on your use case
  - A stricter, safer α means a higher chance of not rejecting H_0 although we should have
  - A higher, more relaxed α means a higher chance of rejecting H_0 when H_0 is true (=we shouldn't)
  - Sample size is important, as it increases power (the chances of finding a difference if there is one)

Questions?

PRELIMINARY COURSE PLAN

- Introduction, Descriptive statistics: measures of central tendency
- Introduction to R, Descriptive statistics: measures of dispersion, distributions, visualizations
- Basic probability theory
- Introduction to inferential statistics: sampling, hypothesis testing
- Test Statistics (I)
- No class *(Tag der Lehre)*
- [**Today** - Nov 27th] Test statistics (II-III)
- [Dec 4th] Introduction to modeling: correlation
- [Dec 11th-18th] The Generalized Linear Model
- [Jan 9th-29th] Extra session if needed, Mock Exam, QA, extra exercises

TODAY'S TOPICS AND LEARNING GOALS



↗ Test Statistics

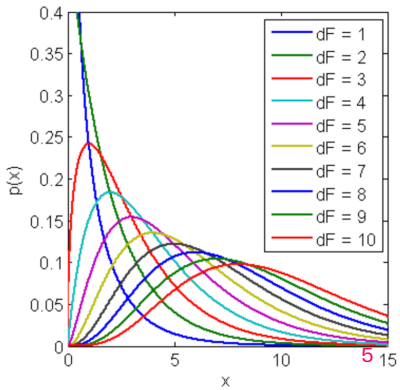
- ↗ Understand and explain what a **test statistic** is and what it is used for

↗ Student's t-test

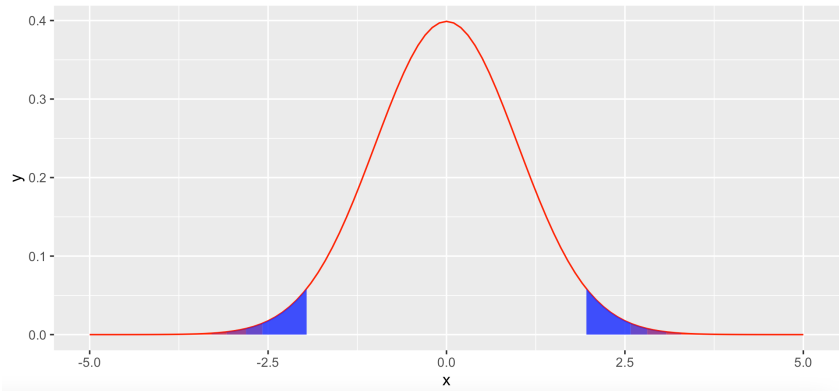
- ↗ Compute the **one-sample t-test**, the **two-sample independent t-test** and the **two-sample paired t-test** and identify when to use them

↗ The Chi-Square (χ^2) test

- ↗ Compute the **Chi-Square (χ^2) Goodness-of-fit test** and the **Test of Independence** and identify when to use them



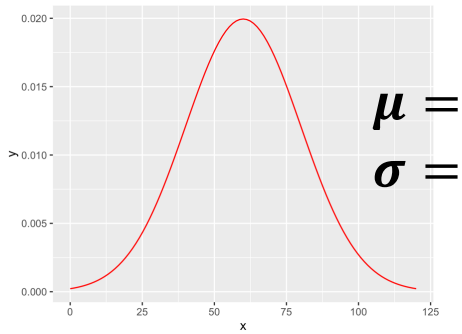
TEST STATISTICS



Howell, Fundamental Statistics for the Behavioral Sciences

USING THE SAMPLING DISTRIBUTION FOR HYPOTHESIS TESTING

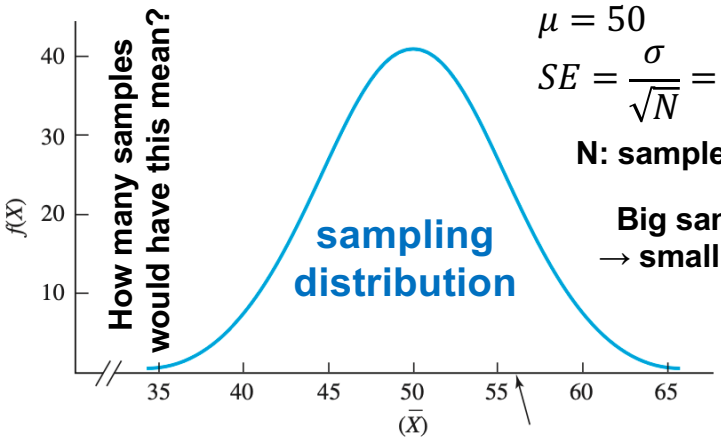
Mean of a sample



$\mu = 50$
$\sigma = 10$

1. How does the population of non-stressed children look like?
   We know the size and mean of the population
2. We know the size and the mean of our sample
3. How likely is it that, if we picked **a random sample** from the population, their mean would be ***as high as 56?***

$N = 5$
$\bar{X} = 56$



$\mu = 50$

$SE = \frac{\sigma}{\sqrt{N}} = 4.47$

N: sample size

Big sample size → small variation!

mean of behavior issue scores in a sample

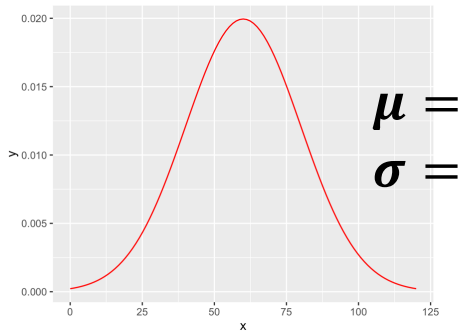
- It's the area below the curve for $\bar{X} = 56$ in the **sampling distribution**
- It's the same as computing the area below the curve in the **sampling distribution of z-scores**

$$z_{\bar{X}} = \frac{\text{score} - \text{mean}}{SE} = \pm 1.96$$

Howell, Fundamental Statistics for the Behavioral Sciences

USING THE SAMPLING DISTRIBUTION FOR HYPOTHESIS TESTING

Mean of a sample



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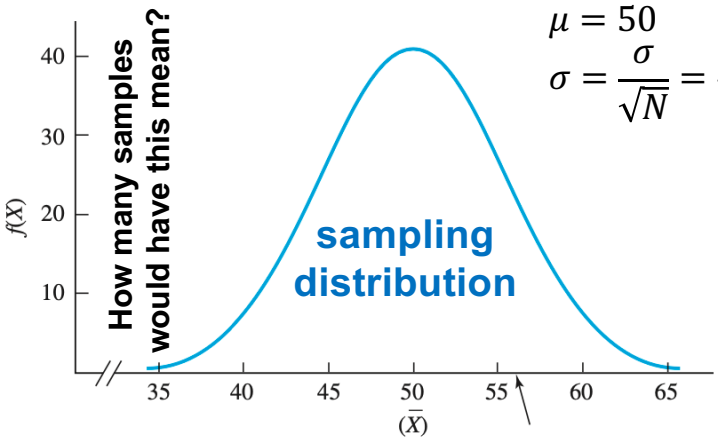


$\mu = 50$
$\sigma = \frac{\sigma}{\sqrt{N}} = 4.47$

mean of behavior issue scores in a sample

$\alpha = 0.05$

- ↗ **low probability** ($p < \alpha$): reject H_0
- ↗ **not so low probability** ($p \geq \alpha$): we fail to reject H_0

$P(\text{sample}|H_0) = 0.0901$

We can't reject the Null Hypothesis!

TEST STATISTICS

- The amount of variance is very important when we judge if a difference is a ***real effect***, something we care about or if it's ***random***
  - *Is a difference in 10000 euros yearly income a big difference? Or do we expect this difference in the distribution under H_0?*

$$\text{test_statistics} = \frac{\text{observed_data} - \text{what_we_expect_under_}H_0}{\text{average variation}}$$

- We use **test statistics** to
  1. look at the difference between **what we see in the data** and **what we'd expect to see** if H_0 is true
  2. compare this difference with **the "average" variation** we'd expect to see

THE Z-SCORE AS A TEST STATISTIC FOR A NORMAL DISTRIBUTION

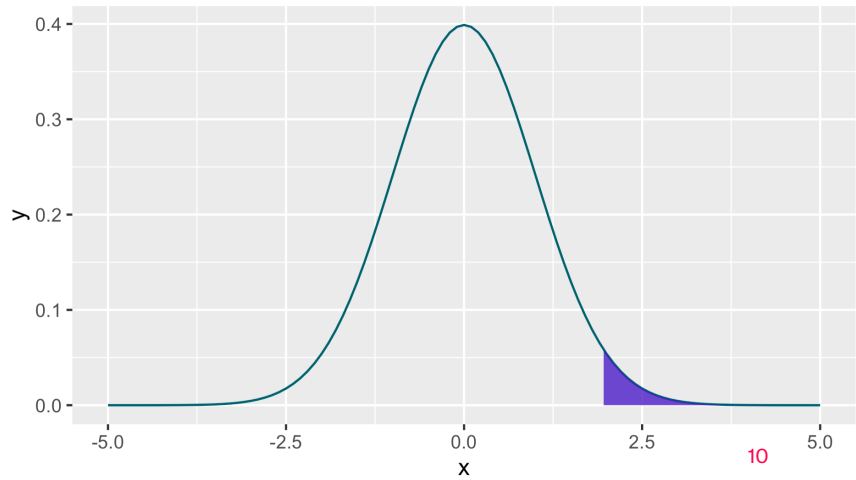
$$\text{test_statistics} = \frac{\text{observed_data} - \text{what_we_expect_under_}H_0}{\text{average variation}}$$

- A z-score is a test statistic, the **z-statistic**:

$$z = \frac{\text{score} - \text{mean}}{\text{SD}} = \frac{X - \mu}{\sigma_X}$$

  - using μ and σ we find out **how much a score differs from the mean** and how this variation relates with the **variation we expect**
    - we learn where (how likely) we would expect a score to be in the distribution

Distribution of z-scores
Standard Normal Distribution



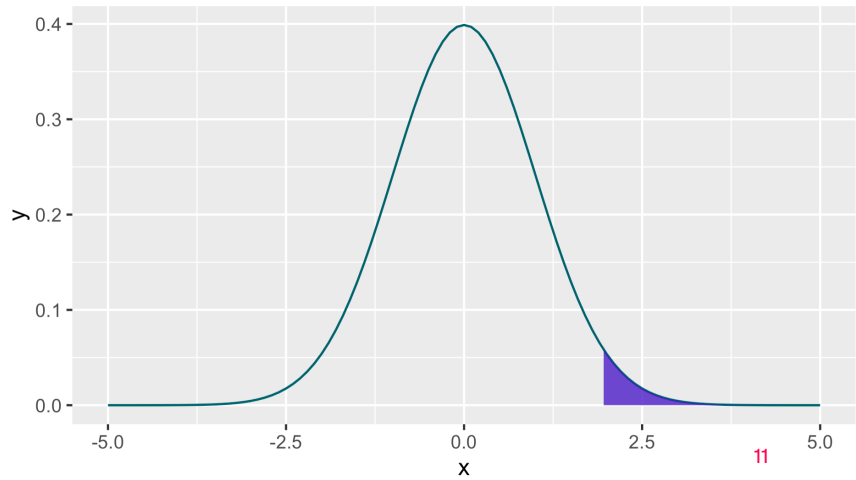
THE Z-SCORE AS A TEST STATISTIC FOR A NORMAL DISTRIBUTION

$$\text{test_statistics} = \frac{\text{observed_data} - \text{what_we_expect_under_}H_0}{\text{average variation}}$$

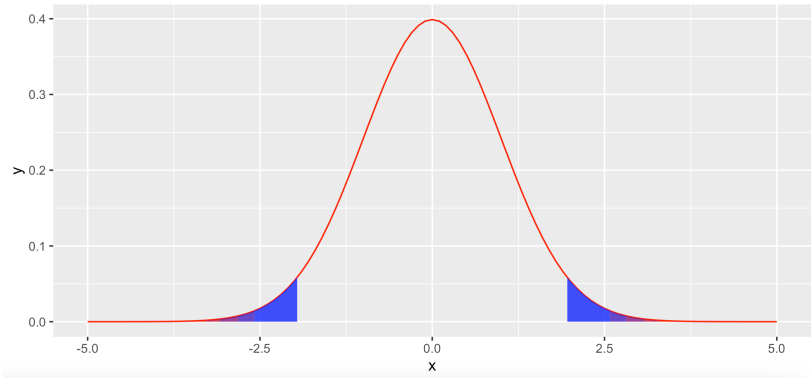
- **z-statistic**: for a sample:

$$z = \frac{\text{sample_stat} - \text{mean_stat}}{\text{SE}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{N}}}$$

  - we find out **how much a sample statistic differs from the mean population statistic** and how this variation relates with the **variation we expect in the sampling distribution**
  - we learn where (how likely) we would expect this sample to be in the sampling distribution



FROM TEST STATISTICS TO STATISTICAL SIGNIFICANCE



↗ Given a certain statistics, we can determine if it provides strong enough evidence to reject the H_0

1. We set the α for the test, e.g. 0.05 (and check if it's a one- or two-tailed test)
2. **z-test:** we check in the **z-distribution** if the **z score** for our sample is below or beyond a **critical cutoff value**, that is if it is ***extreme enough*** to *reject the* H_0

	"significant"		"highly significant"		
α	**0.05**	**0.01**	**0.005**	**0.002**	**0.001**
z_α for 1-Tailed Tests	**±1.645**	**±2.33**	**±2.58**	**±2.88**	**±3.08**
z_α for 2-Tailed Tests	**±1.96**	**±2.58**	**±2.81**	**±3.08**	**±3.29**

R_exercise_14.R

Questions?

STUDENT'S T-TEST

STUDENT'S T-DISTRIBUTION

$$z = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{N}}} = \frac{\bar{X} - \mu}{\sqrt{\frac{\sigma^2}{N}}}$$

$$t = \frac{\bar{X} - \mu}{s_{\bar{X}}} = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{N}}} = \frac{\bar{X} - \mu}{\sqrt{\frac{s^2}{N}}}$$

- What do we do if σ is unknown?
  - We have to estimate the expected variation by way of the sample SD (s) instead of the population SD σ
  - We need to use a different distribution: **Student's t-distribution**
  - Corresponding test: the **t-test**
    we check in the **t-distribution** if the **t** for our sample is below or beyond a **critical value**

William S. Gosset (aka *Student*) statistician, chemist and brewer (hired by Guinness to conduct studies on barley to brew beer)

STUDENT'S T-DISTRIBUTION

- **William S. Gosset (aka *Student*)** was a statistician, chemist and brewer, hired by Guinness
- He conducted studies on the chemical properties of barley (barley quality)
  - He wanted to ***compare*** samples to a population (and samples with other samples)
  - He was working with small samples, so he wanted to know how much he could ***rely on*** a small sample

Does my sample have the same quality as the Guinness should always have?

How do I deal with small samples?

STUDENT'S T-DISTRIBUTION

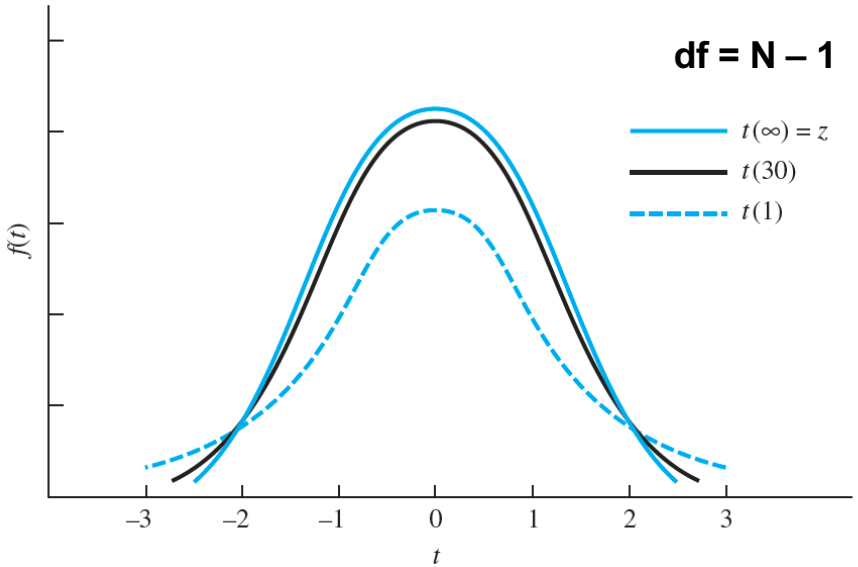
$$t = \frac{\bar{X} - \mu}{s_{\bar{X}}} = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{N}}} = \frac{\bar{X} - \mu}{\sqrt{\frac{s^2}{N}}}$$

- We need to use a t-distribution
  - ***similar*** to the z-distribution ***but with thicker tails***
    (= **more uncertainty** extreme values will be more common)
  - as the sample size grows, the t-distribution will **converge to the z-distribution**
  - If we used the z-distribution (pnorm), it would give us too many false positives

We don't always use the same t-distribution

It depends on the size of the sample (N)!

Or more specifically on the Degrees of Freedom (df)



THE ONE-SAMPLE T-TEST

Example 1 (two-tailed hypothesis)

$\mu = 330$

$N = 20$
$\bar{X} = 328$
$SD = 4$

- There's a company producing energy drinks in 330 ml cans
- We take a sample of 20 bottles: are they significantly different than what the company should sell (330 ml cans)?
  - H_1: sample comes from a different population (with < 330ml or > 330 ml in the cans)
  - H_0: sample comes from the population of "standard" 330 ml can

(two-tailed)

$\alpha = 0.05$

1. Compute ***t***

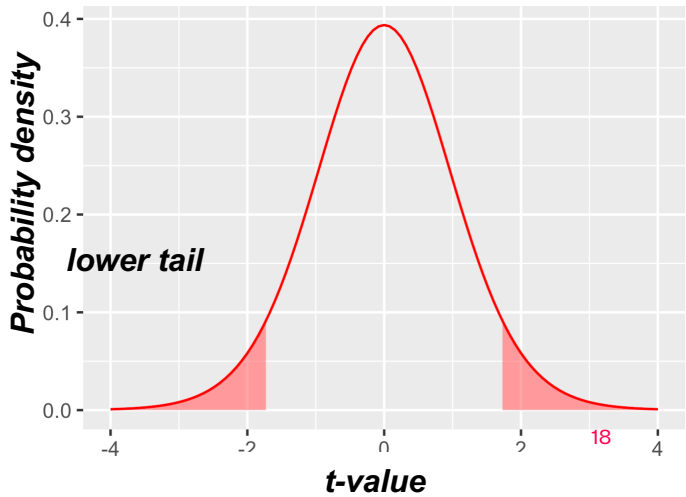
$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{N}}} = \frac{328 - 330}{\frac{4}{\sqrt{20}}} = -\frac{2}{0.8944} = -2.236$$

2. Check the area under the curve for the ***t-distribution*** for a value as extreme as ***t***

```
2*pt(-2.236, df=19, lower.tail=TRUE)
```

2*pt because we had a two-tailed hypothesis — ***t*** — ***N-1*** — **we want the lower tail (lower.tail = FALSE is analogous to 1-pnorm for the z-test)**

The t-distribution



$P(\text{sample}|H_0) = 0.0375$

We can reject H_0!

THE ONE-SAMPLE T-TEST

Example 2 (one-tailed hypothesis)

$\mu = 3.87$

$N = 36$
$\bar{X} = 4.39$
$SD = 2.61$

- The Lie Scale score measures how much children lie, and for elementary school children, it has a population mean of 3.87
  - Sample of 36 children who are going through a stressful time, mean Lie Scale score for sample: 4.39 (SD: 2.61)
  - H_1: this is a "particular" group of children, who lie more because of stress **(one-tailed)**
  - H_0: this is a "standard" group of children

$\alpha = 0.05$

1. Compute ***t***

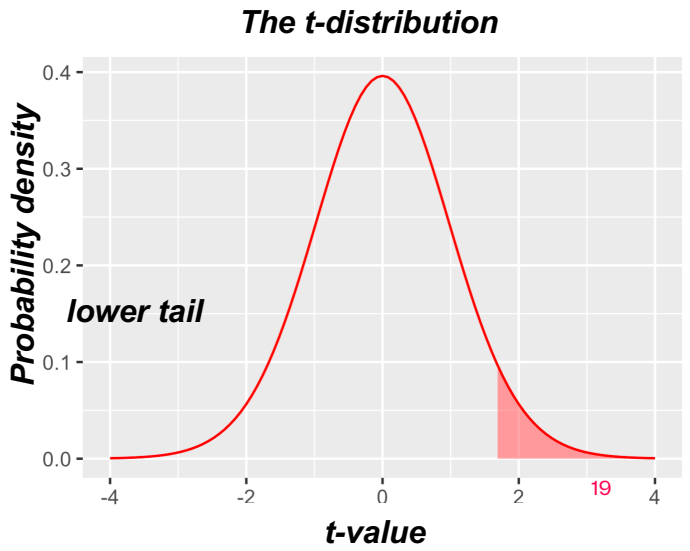
$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{N}}} = \frac{4.39 - 3.87}{\frac{2.61}{\sqrt{36}}} = \frac{0.52}{0.435} = 1.1954$$

2. Check the area under the curve for the ***t-distribution*** for a value as extreme as ***t***

```
pt(1.1954, df=35, lower.tail=FALSE)
```

$P(\text{sample}|H_0) = 0.12$

We can't reject H_0!



THE ONE-SAMPLE T-TEST

Example 3 (using a dataset in R)

$\mu = 70.8$

$N = 9$
$\bar{X} = 73$
$SD = 9.25$

(two-tailed)

$\alpha = 0.05$

- If we know that the average height of the European adults is 70.8 inches
  - we take 9 students from the dataset `student.csv`
  - H_1: this is a "particular" group of students, who are taller or shorter than the population
  - H_0: this is a "standard" group of students
  - We can solve this like the previous examples
  - Or, as we can load the dataset in R, we can use the t.test() function directly:

```
t.test(x = samplevector, mu = 70.8,
       alternative = 'two.sided')
```

```
        One Sample t-test

data:  samplevector
t = 0.72802, df = 8, p-value = 0.4874
alternative hypothesis: true mean is not equal to 70.8
95 percent confidence interval:
 65.93519 80.15370
sample estimates:
mean of x
 73.04444
```

We can't reject H_0!

DEGREES OF FREEDOM

- Imagine you go out to drink beer with your friends, **Sarah** and **William**
- Your roommate asks how many beers you drank, and you say ***"on average, 3 each"***
- If your roommate asks **Sarah**, and knows that **Sarah** had **1 beer** (**n1**), can they figure out how many beers you had (**n3**)?
  - What if they know also that **William** had **3 beers** (**n2**)?

$$n3 = (\text{mean} * 3) - n1 - n2$$

$$n3 = 9 - 1 - 3 = 5$$

- If you only know the mean, **n1**, **n2** and **n3** are ***free*** to be any number
  - Once you know **N-1** scores (**n1**, **n2**), **n3** is not ***free*** anymore
  - **n1**, **n2** are ***independent*** pieces of information (**n3** will be dependent)

DEGREES OF FREEDOM

- Degrees of freedom are the ***number of independent pieces of information we need*** to make a calculation
  - for **n3** in our example, you need the mean, **n1** and **n2**
  - for the **mean**, if you don't know it, you need **n1**, **n2**, **n3**
  - what about the SD?

$$s_x = \sqrt{\frac{\sum(X - \bar{X})^2}{N - 1}}$$

The mean is already part of the SD equation (1 df)!

We can take 1 df out

  - To know the SD, you must know the mean
  - If you know the mean, you don't need to know all and each value in the sample

COMPARING TWO SAMPLES: THE TWO-SAMPLE T-TEST

- So far we have drawn connections between **one sample** and a population

- But what if we want to compare **two samples**?
  - E.g. how people rate a website in two different versions (AB testing)
  - ... if we find a difference, is it enough to say one is “better”?

Do two groups of beer (fermented in different conditions) have different acidity?

THE TWO-SAMPLE INDEPENDENT T-TEST

- H_0: the difference between the groups is due to chance ($\mu_A - \mu_B = 0$)
- H_1: the difference between the groups is not due to chance ($\mu_A - \mu_B \neq 0$)

$$\text{test_statistics} = \frac{\text{observed_difference} - \text{what_we_expect_under_}H_0}{\text{average variation}}$$

$$t = \frac{(\bar{X}_A - \bar{X}_B) - (\mu_A - \mu_B)}{\sqrt{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)}} = \frac{(\bar{X}_A - \bar{X}_B) - 0}{\sqrt{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)}}$$

the sample variance needs to take into account that we have 2 samples

1. Compute test statistic ***t***
2. Compare ***t*** with the **t-distribution**: how extreme would such a ***t*** be under H_0 (how extreme would such a difference be?)

THE TWO-SAMPLE INDEPENDENT T-TEST

Example

♂ $N = 50$
$\bar{X} = 46050$
$SD = 10303$

♀ $N = 50$
$\bar{X} = 37375$
$SD = 7396$

- Imagine you have two samples of graduates (men and women) together with their income data
  - H_1: there is a difference between the groups in the population, which is due to gender ($\mu_A - \mu_B \neq 0$)
  - H_0: there is no difference between the groups in the population, the difference in the sample is due to chance ($\mu_A - \mu_B = 0$)

1. Compute test statistic ***t***

$$t = \frac{(\bar{X}_A - \bar{X}_B) - (\mu_A - \mu_B)}{\sqrt{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)}} = \frac{(\bar{X}_A - \bar{X}_B) - 0}{\sqrt{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)}}$$

and compare ***t*** with the **t-distribution** using the `pt()` function in R

2. (or) run the test directly in R

```
t.test(male.sample, female.sample,
paired = FALSE, alternative = "two.sided",
var.equal = TRUE)
```

"two.sided" if your H_1 is two-sided,
"less" if your H_1 is that $\mu_A < \mu_B$,
"greater" if your H_1 is that $\mu_A > \mu_B$,

COMPARING TWO SAMPLES: THE TWO-SAMPLE PAIRED T-TEST

- Imagine you worked on a new version of the THA website and you want to know which version is better
- You assigned people to the A or B version of the website randomly – good!
  - But.. it's useful information ***who*** gave the rating!
    - Some people have a tendency for higher ratings
    - Some people have a tendency for lower ratings
- You could try to show both versions to all participants!

COMPARING TWO SAMPLES: THE TWO-SAMPLE PAIRED T-TEST

- Some of the variation in our ratings is due to the fact that people have their own behavior
  - We want to remove "individual variance" from our data
  - Let's just look at the differences for the two versions for each person

	Version A	Version B	difference
Dustin	6.3	6.4	-0.1
Jane	6.8	7	-0.2
Jonathan	6.1	6.5	-0,4
Lucas	8	8	0
Max	9	9.6	-0.6
Mike	5.9	6	-0.1
Nancy	6.5	6.8	-0.3
Steve	5.3	5.8	-0.3
Will	7.3	7.4	-0.1

COMPARING TWO SAMPLES: THE TWO-SAMPLE PAIRED T-TEST

- H_1: there is a difference between the groups, which is due to one version being better ($\mu_A - \mu_B \neq 0$)
- H_0: the difference in the sample is due to chance ($\mu_A - \mu_B = 0$)

$$\text{test_statistics} = \frac{\text{observed_difference} - \text{what_we_expect_under_}H_0}{\text{average variation}}$$

$$t = \frac{\bar{D} - 0}{s_{\bar{D}}} = \frac{\bar{D} - 0}{\frac{s_D}{\sqrt{N}}}$$

We estimate the variance by way of the sample *s* (SD of the mean difference)

	Version A	Version B	difference
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Nancy	6.5	6.8	-0.3
Steve	5.3	5.8	-0.3
Will	7.3	7.4	-0.1

In R:

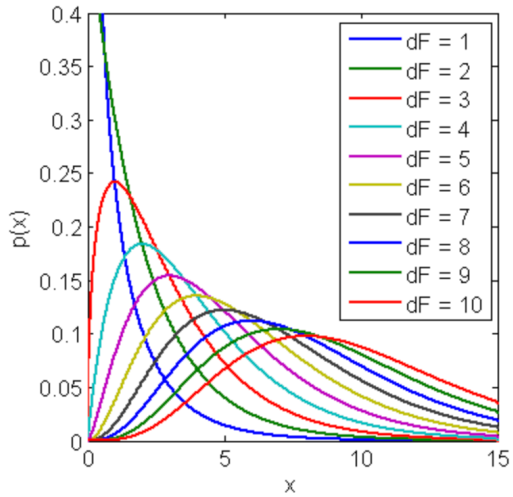
```
t.test(scores_A, scores_B, paired = TRUE,
       alternative = "two-sided")
```

"two.sided" if your H_1 is two-sided,
"less" if your H_1 is that $\mu_A < \mu_B$,
"greater" if your H_1 is that $\mu_A < \mu_B$,

R_exercise_15.R

Questions?

THE CHI-SQUARE (χ^2) TEST



CATEGORICAL DATA

- **Categorical** or **discrete** variables can take only a few possible values

  grammatical gender, eye color, voted party

  - **Binary** variables can only take one out of two values

    positive / negative to covid, married / not married

- So far, we have looked at tests where the data is continuous
  (can take any value between the highest and lowest point in the scale)

How does test statistics work for categorical data?

CATEGORICAL DATA

	In favor	Opposed	total
observed	58	42	**100**
H_0	50	50	**100**

- Imagine we run a survey on the use of ChatGPT in class.
  We want to know if teachers are **in favor** or **opposed to** it
  - We ask 100 teachers
  - Are significantly more teachers in favor or opposed?
- Do these frequencies represent a chance deviation from a random 50/50 split?
  - The test we use in this case is the **Chi-Square (χ^2) Goodness-of-fit Test**

THE CHI-SQUARE (χ^2) GOODNESS-OF-FIT TEST

$$\text{test_statistics} = \frac{\text{observed_data} - \text{what_we_expect_under_}H_0}{\text{average variation}}$$

- The Chi-Square (χ^2) goodness-of-fit test compares ***observed frequencies*** with ***expected frequencies***

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

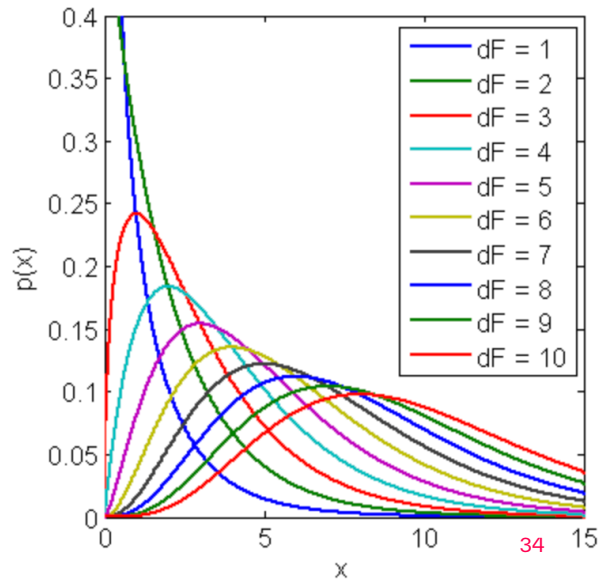
- How large is the difference between ***observed*** and ***expected frequencies***?
  - It depends on **how large a number we expect**

- Why squared? Because otherwise the $O - E$ differences would cancel each other out
- Why a sum? Because we need to do this over all categories

THE CHI-SQUARE (χ^2) GOODNESS-OF-FIT TEST

	In favor	Opposed	total
observed	58	42	**100**
H_0	50	50	**100**

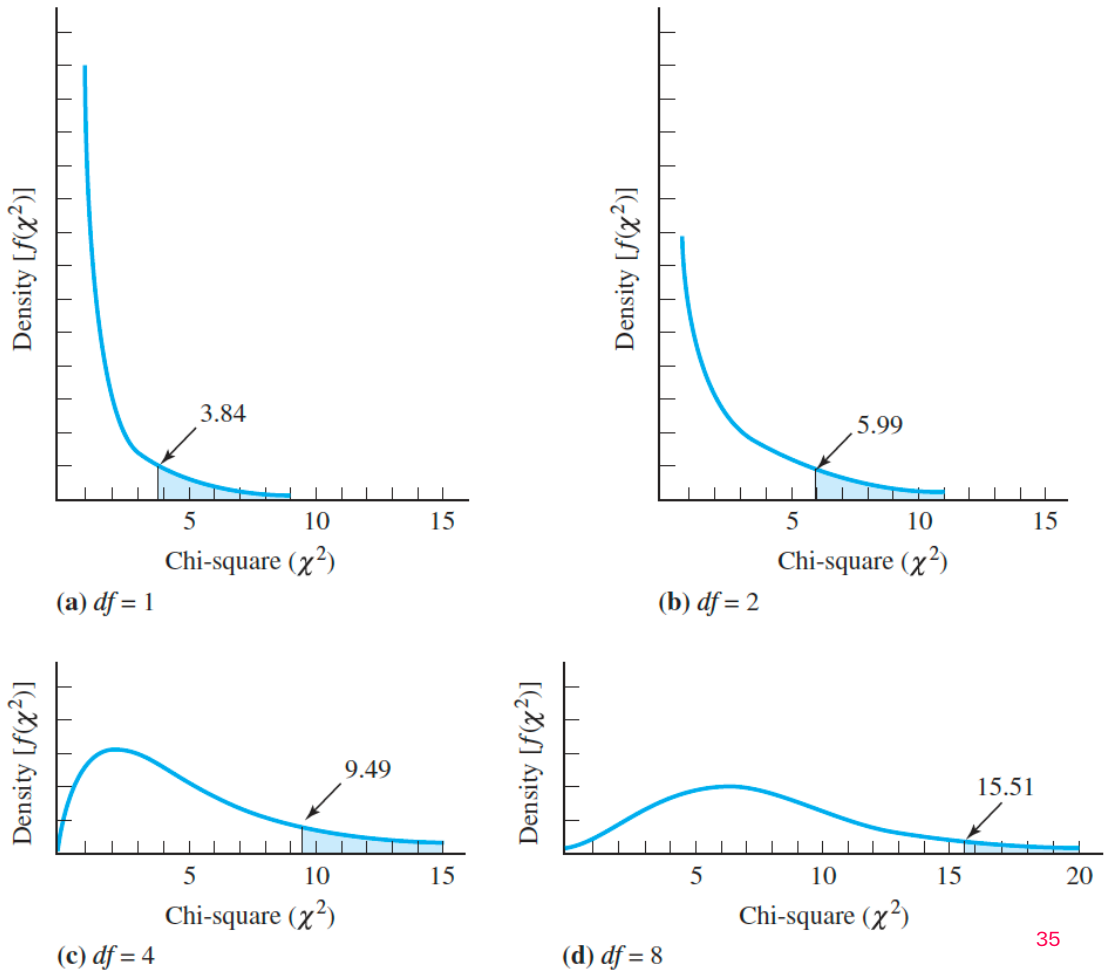
- **Test statistics:** $\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(58-50)^2}{50} + \frac{(42-50)^2}{50} = \frac{64}{50} + \frac{64}{50} = 2.56$
- Is this value large enough to reject the H_0 for $\alpha = 0.05$?
  - We need to compare the obtained χ^2 with the **distribution of χ^2** to evaluate if the χ^2 we obtained is an extreme value
- Like the t-distribution, also the **χ^2 distribution** has a different shape depending on the degrees of freedom
  - df = n_categories − 1



Howell, Fundamental Statistics for the Behavioral Sciences

THE CHI-SQUARE (χ^2) TEST

- Different χ^2 distributions for different df
- For each distribution, the arrow indicates the **cutoff value** delimiting extreme values for $\alpha = 0.05$



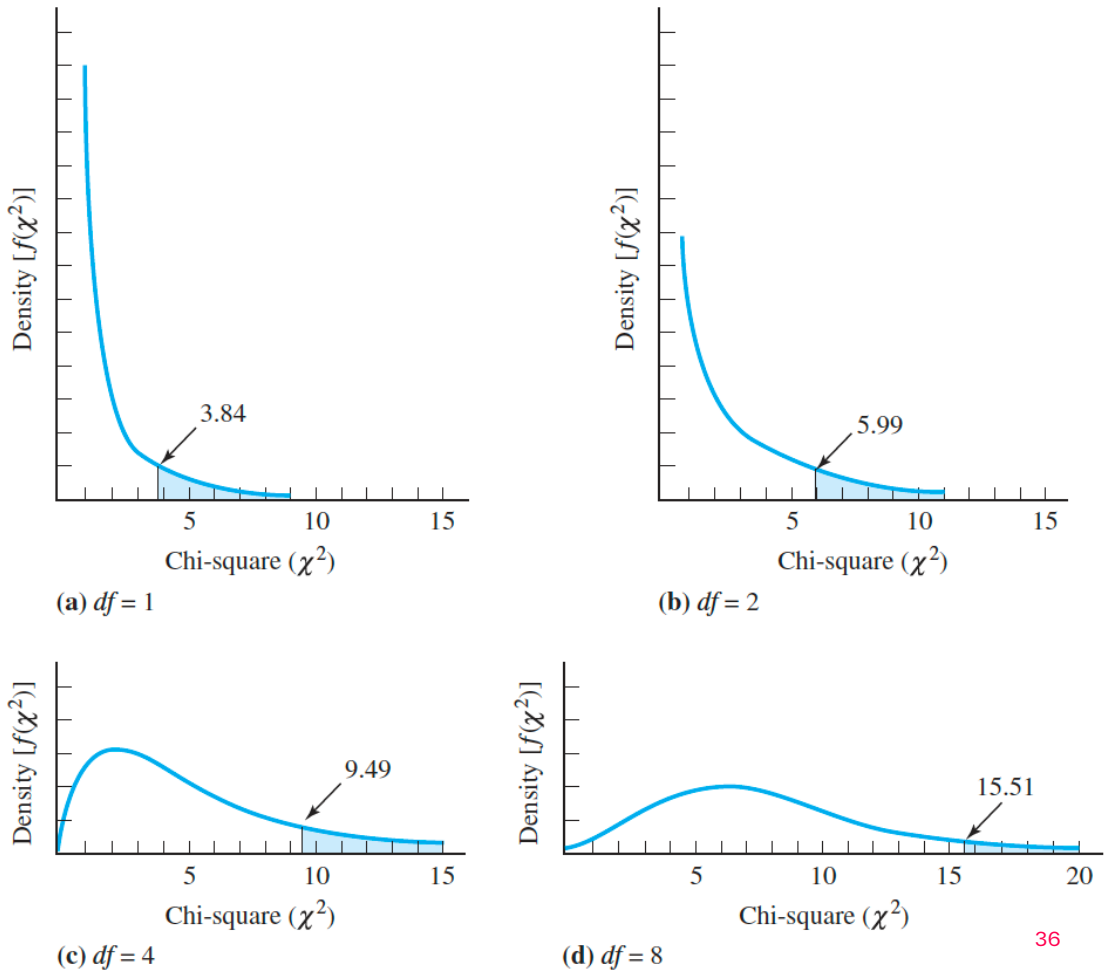
Howell, Fundamental Statistics for the Behavioral Sciences

THE CHI-SQUARE (χ^2) TEST

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

The χ^2 test is **always one-tailed**

- we reject H_0 for large values of χ^2, not for small ones
- if we have 2 categories though, we would get a large χ^2 no matter which one is much bigger than chance



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THE CHI-SQUARE (χ^2) TEST

$\alpha = 0.05$

For our $\chi^2 = 2.56$ with df $= 1$

- If you know it, you can check the cutoff for $\alpha = 0.05$, which is $\chi^2 = 3.84$
  - our χ^2 is smaller than that, so we can't reject H_0
- Or you can calculate the area under the curve with R:

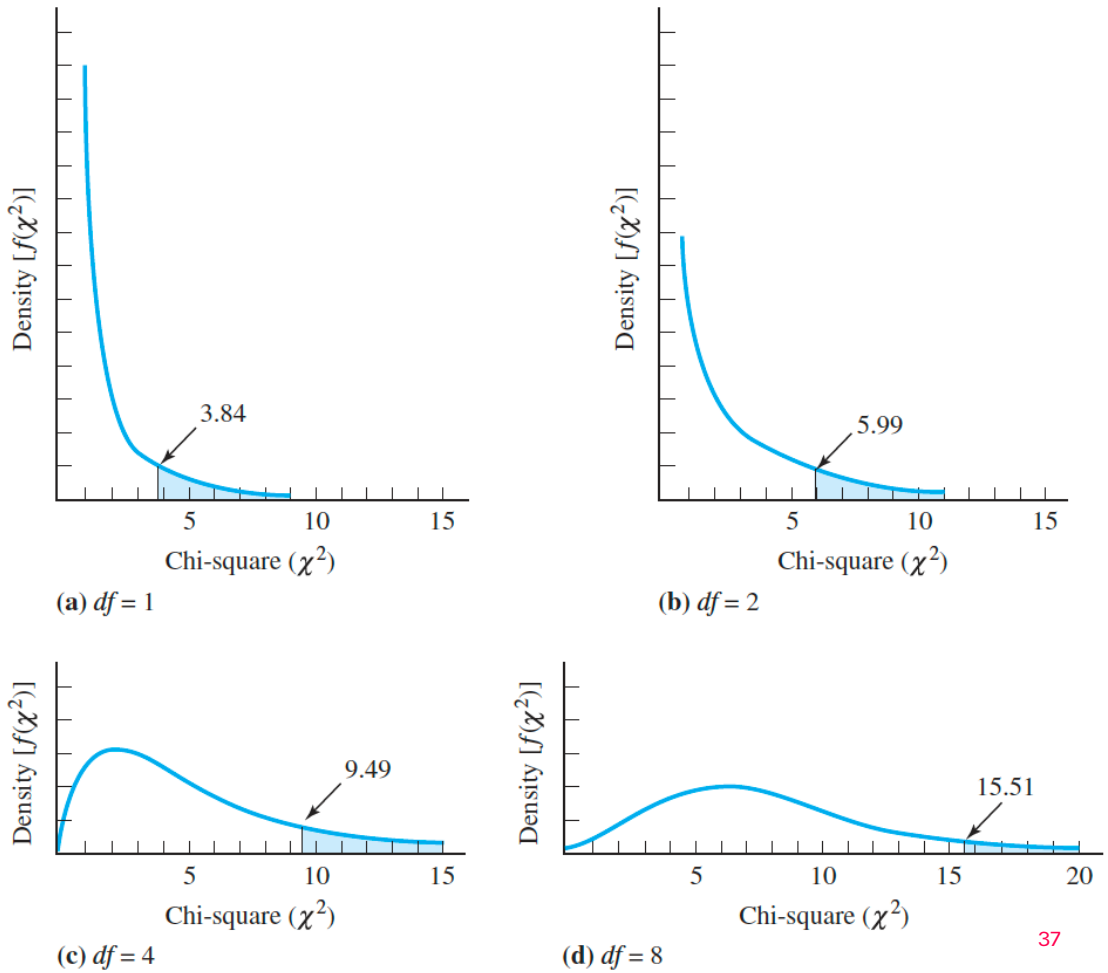
```
1- pchisq(2.56, df=1)
```

or

```
pchisq(2.56, df=1, lower.tail = FALSE)
```

$p = 0.89$

We can't reject H_0!



THE TEST OF INDEPENDENCE

Type of Fries	Northeast	South	Central	West	
skinny fries	70	50	20	25	165
curly fries	100	60	15	30	205
	170	110	35	55	370

- If we have a contingency table (each observation classified on the basis of 2 variables) we need to do another version of the χ^2 test: a **Test of Indipendence**
- The **Test of Indipendence** checks that being a member of one category (fries preference) is ***independent*** from being the member of another (which part of the US one comes from)
  - H_1: the fries preference depends on where you are from
  - H_0: the proportions of skinny vs. curly fries are the same no matter where you are from

THE TEST OF INDEPENDENCE

Type of Fries	Northeast	South	Central	West	
skinny fries	70	50	20	25	165
curly fries	100	60	15	30	205
	45.9%	29.7%	9.5%	14.9%	370

H_0: same fries preference for both kinds of fries!

Type of Fries	Northeast	South	Central	West	
skinny fries	165x45.9%	165x29.7%	165x9.5%	165x14.9%	165
curly fries	205x45.9%	205x29.7%	205x9.5%	205x14.9%	205
	45.9%	29.7%	9.5%	14.9%	370

THE TEST OF INDEPENDENCE

Observed

Type of Fries	Northeast	South	Central	West	
skinny fries	70	50	20	25	165
curly fries	100	60	15	30	205
	170	110	35	55	370

H_0: same fries preference for both kinds of fries!

Expected

Type of Fries	Northeast	South	Central	West	
skinny fries	75.81	49.05	15.61	24.53	165
curly fries	94.19	60.95	19.39	30.47	205
	170	110	35	55	370

Now we can use our formula: $\chi^2 = \sum \frac{(O - E)^2}{E}$

THE TEST OF INDEPENDENCE

Observed - Expected

Type of Fries	Northeast	South	Central	West
skinny fries	70-75.81	50-49.05	20-15.61	25-24.53
curly fries	100-94.19	60-60.95	15-19.39	30-30.47

$\alpha = 0.05$

↗ **Test statistics:** $\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{33.77}{75.81} + \frac{33.77}{94.19} + \frac{0.89}{49.05} + \frac{0.89}{60.95} + \frac{19.29}{15.61} + \frac{19.29}{19.39} + \frac{0.22}{24.53} + \frac{0.22}{30.47} = 3.08$

↗ **p-value:**

```
1 - pchisq(3.08, df=3, lower.tail = TRUE)
```
or
```
pchisq(3.08, df=3, lower.tail = FALSE)
```

$\text{df} = (\text{nrows} - 1)(\text{ncols} - 1) = 1 * 3 = 3$

↗ Or directly:

```
chisq.test(my_data_frame)
```

$\boldsymbol{p = 0.38}$

We can't reject H_0!

R_exercise_16.R

CATEGORICAL DATA WITH SMALL SAMPLE SIZES

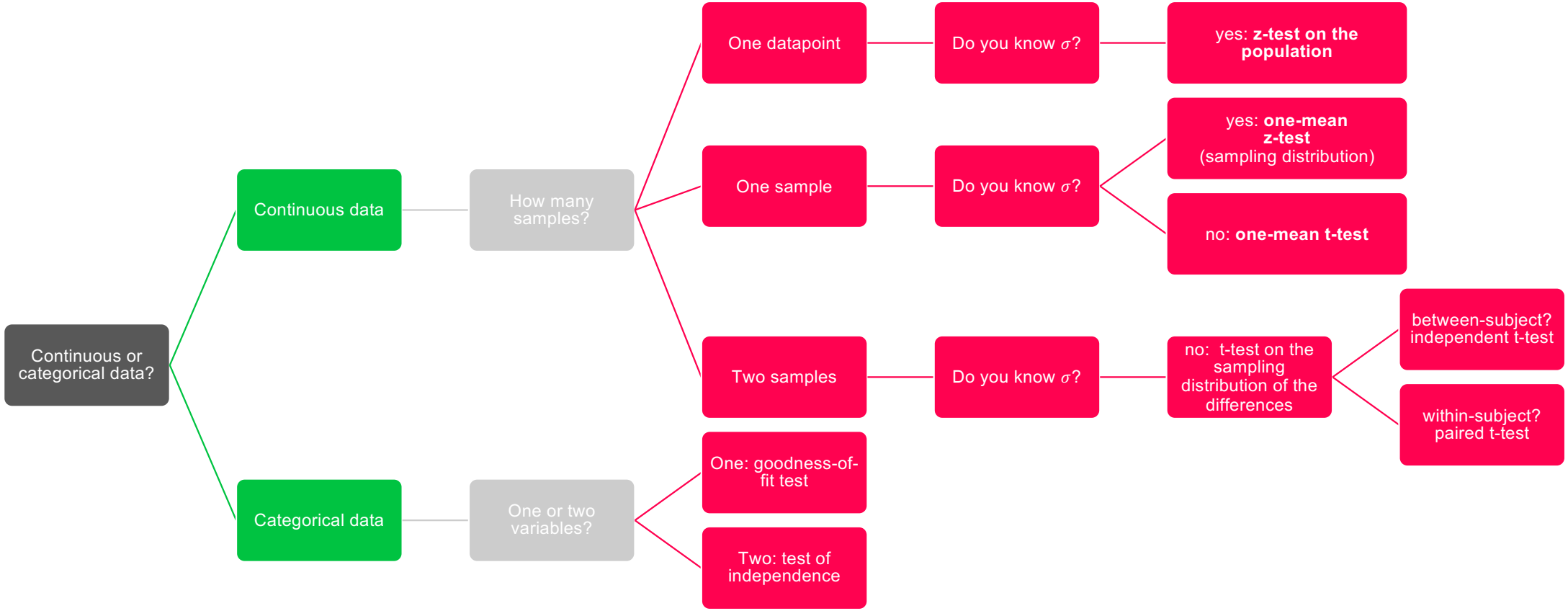
- If the ***expected frequency*** for any cell is smaller than 5, it's not a good idea to do a χ^2 test
  - the sample size is too small, the sampling distribution of the test statistic is too deviant from a χ^2 distribution

Alternative:

- **Fisher's exact test**
  - normally used on 2 × 2 contingency tables (i.e., two variables each with two options) and with small samples
    (not for the exam but FYI)

Questions?

SUMMING UP


Continuous or categorical data?
Continuous data
How many samples?
One datapoint
Do you know σ?
yes: **z-test on the population**
One sample
Do you know σ?
yes: **one-mean z-test** (sampling distribution)
no: **one-mean t-test**
Two samples
Do you know σ?
no: t-test on the sampling distribution of the differences
between-subject? independent t-test
within-subject? paired t-test
Categorical data
One or two variables?
One: goodness-of-fit test
Two: test of independence

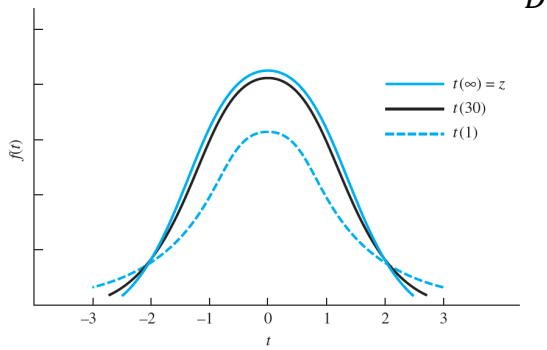
SUMMING UP (1)

So far we looked at "lucky situations" where we know μ and σ

- Usually we are not this lucky, and we do not know σ
  - The **t-distribution** comes to our rescue, allowing us to use s instead

- The following t-tests rely on the t-distribution:
  - **One-sample t-test**: $$t = \frac{\bar{X} - \mu}{s_{\bar{X}}} = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{N}}} = \frac{\bar{X} - \mu}{\sqrt{\frac{s^2}{N}}}$$
  - **Two-sample independent t-test:** $$t = \frac{(\bar{X}_A - \bar{X}_B) - (\mu_A - \mu_B)}{\sqrt{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)}} = \frac{(\bar{X}_A - \bar{X}_B) - 0}{\sqrt{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)}}$$
  - **Two-sample paired t-test:** $$t = \frac{\bar{D} - 0}{s_{\bar{D}}} = \frac{\bar{D} - 0}{\frac{s_D}{\sqrt{N}}}$$

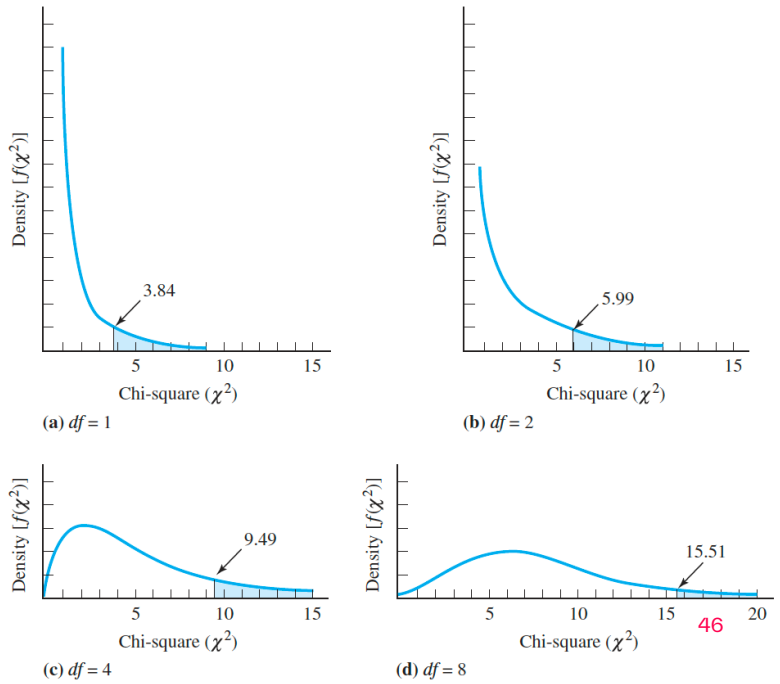
- To know the p-value for a **t-score** we need to know which **t-distribution** to consider: we need to know the degrees of freedom



SUMMING UP (2)

- In case of categorical data, we can test if their frequency distribution differs (represents a change deviation) from the one expected given the Null Hypothesis
  - **One variable: Chi-Square (χ^2) Goodness-of-fit Test**
  - **Two variables: Test of Indipendence**

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

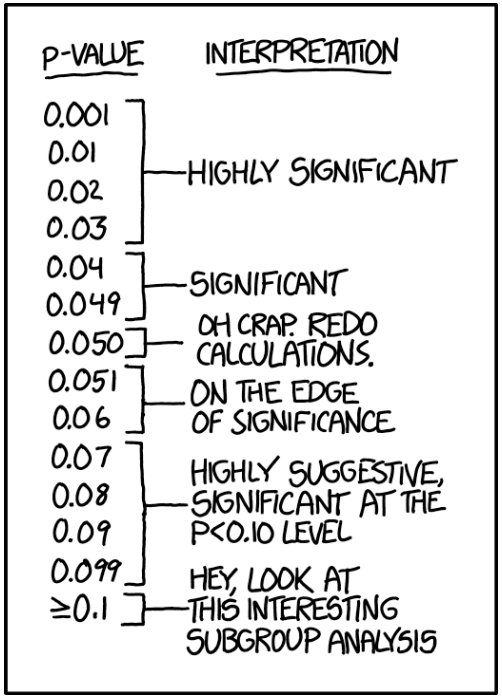


- All these tests can be described by the generic formula for the test statistics

$$\text{test_statistics} = \frac{\text{observed_data} - \text{what_we_expect_under_}H_0}{\text{average variation}}$$

Questions?

HOMEWORK

- Chapter 9, sections 9.6 and 9.7 of “Statistics for Linguists”
- Chapter 12, section 12.3, Chapter 13, sections 13.1 to 13.2, Chapter 14, section 14.1 and Chapter 19, sections 19.1 to 19.2 of Howell, “Fundamental Statistics for the Behavioral Sciences” (available on Moodle)
- Watch this videos:
  - “Test Statistics” (Crash Course Statistics) https://www.youtube.com/watch?v=QZ7kgmhdIwA
  - “What are degrees of freedom?” https://www.youtube.com/watch?v=rATNoxKg1yA
  - “Chi-Square Tests” (Crash Course Statistics) https://www.youtube.com/watch?v=7_cs1YlZoug

Source: xkcd